
Cash Is Not Zero: Auditing a Weight-Centric Quantitative Trading Pipeline

Kevin Lee

University of California at Los Angeles

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kevinlee69720@g.ucla.edu

<https://orcid.org/0009-0004-0388-9260>

Abstract

Quantitative trading systems can produce internally consistent backtests while misstating investment performance through omitted cash returns, unmatched benchmarks, hidden transaction-cost assumptions, or research and execution paths that implement different semantics. We present an audit-first evaluation of QUANT-CORTEX, an open-source, weight-centric research and paper-trading pipeline, and use a fixed six-ETF rotation strategy as a deliberately untuned case study. The audit formalizes residual-cash accounting, next-bar execution, mandatory costs, point-in-time contracts, coherent execution state, and machine-readable provenance. On 2,011 U.S. trading sessions from 2018–2025, residual cash earns an SHV adjusted-close proxy and the strategy is compared with passive returns scaled to the same daily risky exposure. The audited strategy earns a 1.33% net CAGR, a -0.14 cash-excess Sharpe ratio, and a -10.33% maximum drawdown. Before costs, it trails an exposure-matched equal-initial-weight buy-and-hold basket by 3.08 percentage points per year in arithmetic return; a paired circular-block bootstrap gives a 95% interval of [-5.58%, -0.56%]. All four predefined overlay variants trail their matched passive comparator, and the conclusion persists across cost levels, subperiods, and two backtest engines. The contribution is not a new alpha model; it is a reproducible negative result and a concrete protocol for distinguishing software correctness from economic evidence.

1 Introduction

Modern quantitative platforms increasingly unify data processing, feature construction, portfolio allocation, simulation, and broker execution [Yang et al., 2020, Liu et al., 2020, Yang et al., 2026]. This integration reduces friction, but it also makes accounting and interface assumptions load-bearing. A strategy that is 40% invested has a 60% cash position whether or not a backtest models it. A benchmark with 100% risky exposure does not answer whether security selection added value at 40% exposure. Likewise, agreement between a chart and a README does not establish causality, cost realism, or execution safety.

These issues are especially consequential in financial machine learning, where observations are dependent, effective sample sizes are small, and model or strategy selection creates substantial false-discovery risk [White, 2000, Harvey et al., 2016, Bailey and de Prado, 2014, Bailey et al., 2017, Arnott et al., 2019]. Reproducible software is necessary but not sufficient: the reproduced quantity must itself be economically well-defined.

This paper makes three contributions. First, it specifies an auditable return contract for weight-centric pipelines: causal target weights, explicit residual cash, mandatory costs, aligned comparators, and separable asset, cash, and cost contributions. Second, it describes engineering controls that bind

reported results to code, inputs, dependencies, and execution state. Third, it reports a fixed negative-result study of a multi-asset rotation strategy. We report all predefined variants rather than selecting a winner, and separate gross signal quality from net implementability. The purpose is to show what changes when the accounting question is posed correctly, not to claim a profitable trading rule.

2 Related work

Portfolio construction originates from explicit trade-offs between return and risk [Markowitz, 1952]; momentum evidence motivates ranking assets by past returns across both securities and asset classes [Jegadeesh and Titman, 1993, Moskowitz et al., 2012]. The case-study strategy uses these ideas only as a test vehicle. We do not claim that its particular grouping, residualization, or overlays improve on that literature.

Open platforms such as Qlib and FinRL provide modular workflows for financial AI research [Yang et al., 2020, Liu et al., 2020]; FinRL-X emphasizes a common weight interface across research and deployment [Yang et al., 2026]. Our focus is narrower and complementary: the accounting and state invariants that must hold when such an interface is evaluated or sent toward a broker.

Sharpe-ratio inference is sensitive to distributional and serial-dependence assumptions [Lo, 2002]. Data snooping and repeated trials further inflate apparent performance [White, 2000, Bailey and de Prado, 2014, Bailey et al., 2017]. We therefore avoid selecting among variants and use paired block resampling for uncertainty. More broadly, the manifest, locked environment, and executable protocol follow the reproducibility goals described by Pineau et al. [2021].

3 Audit protocol

3.1 Weight, timing, and cash contracts

Let $w_t \in \mathbb{R}^N$ be the risky-asset target decided after observing the close at date t . In the audited convention, the target executes on the first bar strictly after t and begins earning returns only after that execution bar. Thus close-derived signals cannot earn the same close they observe. For the long-only case, $w_{i,t} \geq 0$ and $\sum_i w_{i,t} \leq 1$; the residual cash weight is

$$w_t^c = 1 - \sum_{i=1}^N w_{i,t}. \quad (1)$$

For asset returns $r_{i,t}$ and cash-account return r_t^c , pre-cost portfolio return is

$$r_t^{\text{gross}} = \sum_{i=1}^N w_{i,t-1} r_{i,t} + w_{t-1}^c r_t^c. \quad (2)$$

Missing cash observations fail the run instead of silently becoming zero. The after-cost return is $r_t^{\text{net}} = r_t^{\text{gross}} - c_t$. The baseline uses a symmetric 13 basis-point charge per dollar traded with no commission or tax decomposition. This flat model is a first-order sensitivity assumption, not a market-impact model; large or urgent orders require size- and liquidity-aware execution models such as Almgren and Chriss [2001].

We report nominal CAGR, maximum drawdown, annualized one-way turnover, and cash-excess Sharpe

$$\text{SR}_c = \sqrt{252} \frac{\text{mean}(r_t - r_t^c)}{\text{sd}(r_t - r_t^c)}. \quad (3)$$

An SHV adjusted-close series is used as an investable short-duration cash proxy. It is not a frictionless risk-free asset.

3.2 System controls

The implementation separates data, alpha, portfolio, timing, risk, backtest, execution, and strategy layers under one Python package. Allocation contracts are strict at strategy output and relaxed only

when an overlay intentionally reduces gross exposure. Point-in-time merge guards reject fundamentals that were not yet announced. Optional broker, machine-learning, and provider SDKs remain lazy imports so the offline scientific core is testable without credentials.

Research outputs include the generator hash, Git revision, Python and package versions, input SHA-256 digest, full experiment design, and hashes of every table and figure. Paper-trading state is stored as one versioned snapshot of positions, intents, orders, and metadata; writes are atomic and use optimistic concurrency. An order intent is persisted before submission, and an uncertain submission is not retried automatically. Broker conformance tests instantiate the current Alpaca and Interactive Brokers SDK request types, but authenticated transport and venue behavior remain outside this study.

4 Case study and experimental design

4.1 Fixed strategy

The universe contains QQQ, VGT, GLD, TLT, SPY, and VIG, grouped as growth, real assets, and defensive proxies. On the first session of each week, the strategy ranks groups by a 126-session information ratio relative to QQQ and keeps the top two. Within selected groups, it estimates a one-factor regression on a preceding 126-session window and sums residual returns over a separate 126-session formation window, skipping the latest 21 sessions. Positive scores receive proportional long-only weights, capped at 60% per asset. QQQ is also one of the two growth-group members, so that group's active-return spread contains one zero benchmark leg; this asymmetric product specification is held fixed.

Two exposure overlays are evaluated. A seeded three-state Gaussian-mixture regime model maps QQQ return and realized-volatility features to zero, half, or full exposure. A volatility scaler targets 20 annualized volatility points; no VIX series is supplied, so the experiment uses the documented 21-session QQQ realized-volatility proxy. The full strategy uses both overlays. Three predefined variants remove the regime gate, remove the volatility scaler, or remove both. This audit did not tune parameters or select a variant based on the reported period. The unknown number of configurations explored before the audit prevents a defensible multiple-testing-adjusted discovery claim.

4.2 Data, comparators, and uncertainty

The owner-supplied matrix contains adjusted closes from 2016-01-04 through 2025-12-31, retrieved on 2026-06-16 through yfinance 1.4.1 with `auto_adjust=False`. The yfinance documentation describes the Yahoo Finance API as intended for personal use and directs users to Yahoo's terms of use.¹ The repository owner attests permission to publish the derived aggregate results; the raw matrix is not distributed. Its SHA-256 is:

`efb384a62157e56a0cd8065abf45c1ed07d90ec26c681e5d54d74fe4cb9c55e1`

The first 503 sessions are signal history; evaluation covers 2,011 sessions from 2018-01-02 through 2025-12-31.

Comparators share one capital clock. SPY and a six-ETF basket initialized at equal dollar weights are measured from the close immediately preceding the evaluation window. To isolate cross-sectional decisions from exposure timing, the basket return is multiplied by the strategy's realized risky exposure each day and the remainder earns SHV. This comparator is gross of its own trading costs. We therefore report both gross-to-gross signal comparisons and the more conservative net-strategy-to-gross-benchmark comparison.

For dependent daily observations, we resample paired active-return series with the circular-block bootstrap of Politis and Romano [1992]. We use 5,000 replications, 21-session blocks, and seed 42. Intervals quantify sampling variation conditional on this one historical path, strategy specification, cash proxy, and block length; they do not establish stationarity or future performance.

¹<https://ranaroussi.github.io/yfinance/>

Table 1: Audited accounting over 2018–2025. Sharpe ratios use SHV except for the explicitly zero-cash row, which uses a zero reference rate.

Series	CAGR	Cash-excess Sharpe	Max drawdown
Strategy, after costs and SHV cash	1.33%	-0.14	-10.33%
Strategy, before costs and SHV cash	3.26%	0.15	-8.61%
SHV cash proxy	2.50%	–	-0.45%
Exposure-matched passive basket	6.54%	0.69	-7.97%
Strategy, after costs and zero cash	-0.07%	0.02	-14.22%

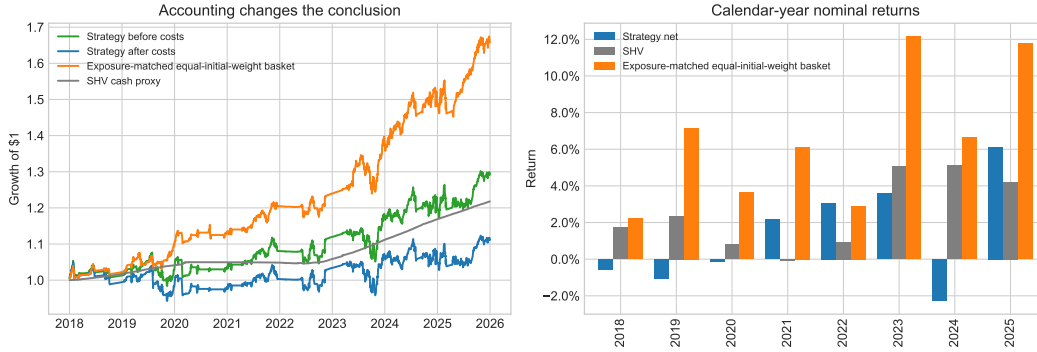


Figure 1: Left: growth of one dollar under audited cash and cost accounting. Right: calendar-year nominal returns. The matched comparator uses the strategy’s daily risky exposure and assigns residual capital to SHV.

5 Results

5.1 Cash accounting changes the reported level, not the verdict

Table 1 and Figure 1 separate cash and cost effects. Crediting residual cash raises net CAGR from -0.07% under a zero-return cash assumption to 1.33%. That correction is economically material because mean risky exposure is only 34.76% and the strategy is fully in cash on 41.92% of sessions. It does not create evidence of alpha: SHV alone compounds at 2.50%, and the exposure-matched passive basket compounds at 6.54%.

5.2 Costs and overlays do not rescue the strategy

The strategy turns over 10.84 times per year on a one-way basis. At zero cost, its cash-excess Sharpe is 0.15; it declines to 0.03 at 5 bps, -0.14 at the 13-bps baseline, -0.41 at 25 bps, and -0.94 at 50 bps. The model is therefore not economically robust to modest execution friction.

Figure 2 compares each variant before costs with a passive basket at the same exposure. This avoids attributing the result solely to strategy turnover. The full model has gross cash-excess Sharpe 0.15 versus 0.69 for its comparator. Removing the regime gate, the volatility scaler, or both increases strategy exposure and raw returns, but every gross strategy still trails its matched comparator. The corresponding annualized gross active returns are -3.08%, -4.63%, -3.21%, and -4.28%. No variant is selected as a replacement model.

5.3 Uncertainty, subperiods, and engine parity

The net strategy’s annualized arithmetic excess return over SHV is -0.94%, with a 95% block-bootstrap interval of [-5.27%, 3.55%]. Before costs, the corresponding estimate is 0.95% [-3.31%, 5.40%]. Neither supports a positive cash-relative claim. Relative to the exposure-matched basket, gross active return is -3.08% [-5.58%, -0.56%], and only 0.64% of bootstrap replications are positive. After costs it is -4.97% [-7.58%, -2.39%], with no positive replication among 5,000 draws.

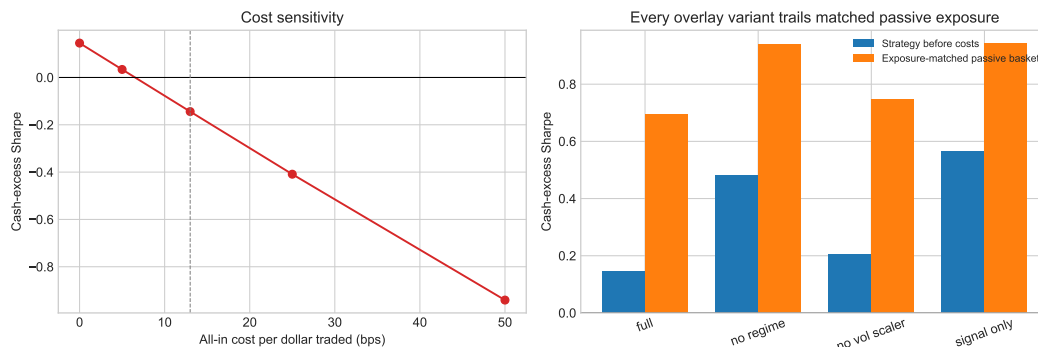


Figure 2: Left: after-cost Sharpe sensitivity for the full strategy. Right: gross-to-gross ablations, so transaction costs do not explain the passive comparator’s advantage. All variants were defined before this comparison.

The conclusion also holds in both prespecified halves. In 2018–2021, strategy CAGR is 0.08% and cash-excess Sharpe is -0.18, versus 4.77% and 0.78 for the matched basket. In 2022–2025, the values are 2.59% and -0.12, versus 8.34% and 0.66. Finally, the vectorized and share-based event-driven engines produce net CAGRs of 1.325% and 1.328%, respectively, and cash-excess Sharpes of -0.1442 and -0.1440. This close agreement is a diagnostic for this experiment, not a claim that their rebalancing semantics are generally equivalent.

6 Limitations and broader impact

This study evaluates one U.S.-listed ETF universe over one eight-year window. SHV has duration, expenses, and market-price variation, so it is only a cash proxy. Adjusted closes omit intraday spreads and fill uncertainty. The flat cost grid excludes order-size, volatility, venue, and market-impact dynamics. The exposure-matched basket is a diagnostic comparator, not a directly quoted security, and is reported gross of its own rebalancing cost. The bootstrap is conditional on a weak-dependence approximation and a chosen 21-session block. Historical research trial counts are unknown, so a Deflated Sharpe Ratio would require an assumed rather than observed number of trials. The data source can revise adjusted histories; the exact input is identified by digest but is not openly redistributed. No live broker authentication, exchange acknowledgment, or funded paper order was tested.

The positive impact of an audit-first protocol is reduced false confidence: cash, costs, timing, and negative results become visible rather than optimized away. The principal risk is the opposite misuse: readers may treat an audited backtest as investment advice or deployment approval. The repository therefore defaults to paper mode, requires pre-trade risk checks, blocks uncertain order retries, and describes the strategy as a research baseline. Past simulated performance does not imply future returns.

7 Conclusion

Software correctness and investment efficacy are different hypotheses. In this case, explicit cash accounting materially improves the reported CAGR, while matched-exposure benchmarks, gross-to-gross ablations, and dependence-aware uncertainty show that the fixed strategy does not add value over simple passive exposure. The broader lesson is procedural: weight-centric trading research should make cash a first-class asset, separate gross signal from net implementation, compare like exposure with like exposure, preserve every predefined variant, and bind artifacts to code and inputs. A negative result under those controls is more useful than a stronger chart built on ambiguous accounting.

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A Reproducibility details

The complete generator is `scripts/run_paper_experiments.py`. From a locked development environment, the fixed command is:

```
PYTHONPATH=. .venv/bin/python scripts/run_paper_experiments.py \  
  --prices-csv local_data/published_rotation_prices.csv \  
  --output-dir paper --bootstrap-replications 5000
```

The input must contain adjusted-close columns QQQ, VGT, GLD, TLT, SPY, VIG, and SHV and match the published digest for exact reproduction. Aggregate CSVs, figures, and their hashes are in `paper/results/manifest.json`. The run used Python 3.14.4 on an eight-core Apple M1 with 8 GiB RAM and completed in approximately 25 seconds on CPU; no GPU was used. Supported CI runs use Python 3.11 through 3.14. The reported experiment consumed less than one CPU-minute; repository-wide testing, packaging, notebook execution, and broker conformance checks were additional engineering validation rather than model search.

B Additional engine comparison

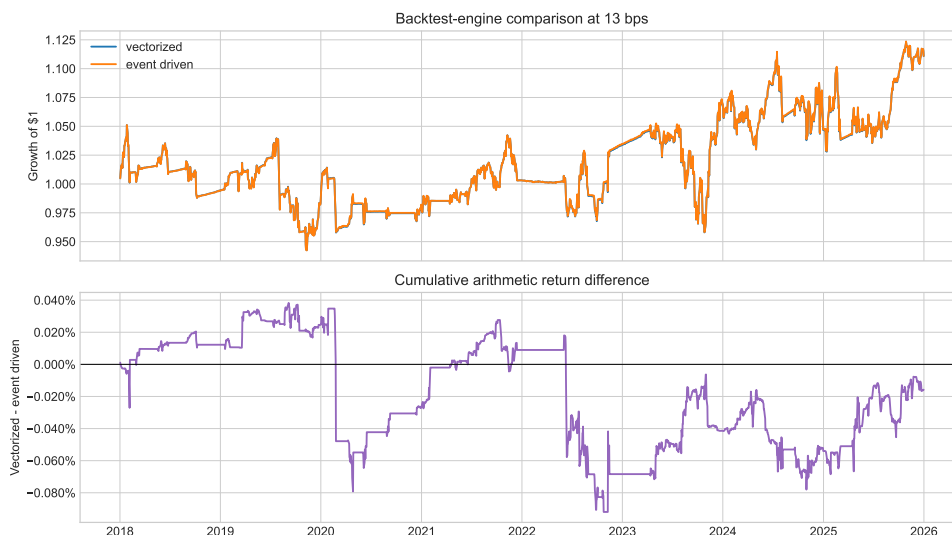


Figure 3: The vectorized engine re-peg target weights between rebalances, whereas the event-driven engine holds explicit drifting shares. Their small difference here does not remove the semantic distinction.

C Use of language-model assistance

An LLM-based coding assistant supported repository inspection, implementation review, regression-test generation, and language editing. It was not a component of the strategy, data, statistical method, or experiment-selection rule. Numerical claims in the paper are generated from committed scripts and machine-readable artifacts; responsibility for the work remains with the author.

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Answer: [No]

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